Empirical Analysis of Tilt-Angle Optimization for Solar Energy Harvesting in Bangalore (12.97°N)

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***Abstract*—Fixed-tilt photovoltaic (PV) installations in Indian urban environments rely predominantly on rule-of-thumb or computationally derived tilt-angle recommendations that are rarely validated by field measurement in tropical cities. This paper presents an empirical investigation of solar panel energy yield at two discrete tilt angles—13° (latitude-matched) and 36° (seasonally steep)—conducted across full solar days in Bengaluru, India (lat. 12.97°N, long. 77.57°E). A multi-sensor data acquisition system comprising an INA219 power monitor, BH1750 ambient light sensor, DS18B20 surface temperature sensor, and MPU-6050 inertial measurement unit was deployed to record voltage, current, irradiance, panel temperature, and measured inclination at one-second intervals. The dataset spans 10,061 one-second samples across six sessions (morning, midday, afternoon) for each tilt angle, collected on 22–23 June 2026. Temperature-corrected power and irradiance-normalised efficiency metrics were derived in post-processing. Our results demonstrate that the 36° configuration produced a cumulative daily energy yield of 0.4626 Wh—a 45.86% improvement over the 0.3172 Wh recorded at 13°—despite operating under lower average ambient irradiance (20,068 lux vs. 30,172 lux). The irradiance-normalised efficiency at 36° was 89.7% higher than at 13°, with average panel temperatures 7.49°C lower. These findings indicate that the conventional latitude-matched tilt angle is sub-optimal for Bengaluru's tropical summer geometry, and underscore the critical role of simultaneous thermal correction in accurate tilt performance assessment.**

***Index Terms*—Solar energy; tilt-angle optimisation; photovoltaic efficiency; Bengaluru; temperature correction; irradiance normalisation; empirical measurement; tropical PV systems.**

I. Introduction

Photovoltaic (PV) technology has emerged as a primary avenue for distributed energy generation in tropical developing economies, where solar irradiance is abundant and off-grid electrification demands are rapidly growing [1]. The annual energy yield of a fixed-tilt PV system is highly sensitive to its inclination angle β: a panel that is poorly oriented relative to the local solar geometry will systematically underperform its rated capacity throughout its operational lifetime [2].

Despite this sensitivity, the vast majority of rooftop solar installations in Indian cities are set at installer-prescribed tilt angles derived from coarse latitude-based heuristics or, at best, from computational models that simulate solar radiation without reference to local atmospheric or thermal conditions [4]. For Bengaluru—a city of over 12 million people at latitude 12.97°N with a distinctive semi-arid tropical microclimate and significant urban heat island (UHI) effects—published empirical tilt-angle data is essentially absent from the literature.

Existing optimisation studies fall broadly into three categories: (i) mathematical regression models applied to meteorological databases [3]; (ii) computational simulations using finite-difference or Liu–Jordan irradiance decomposition techniques [5], [6]; and (iii) tracker-based dynamic systems that are impractical for the large-scale rooftop installations they seek to inform. A comprehensive review by [7] explicitly identifies hardware-based empirical studies in India as a critical gap in the literature. The same review notes that studies specific to Bengaluru (or any South Indian city near the Tropic of Cancer) are absent from the peer-reviewed record.

This paper addresses that gap with four specific contributions:

1) A purpose-built, multi-sensor hardware acquisition system validated for continuous outdoor deployment;
2) An empirically collected dataset of 10,061 one-second-interval samples across two tilt angles and three daily time windows;
3) A thermal derating correction methodology applied in post-processing to isolate geometric efficiency from heat-induced losses;
4) Irradiance-normalised efficiency curves that decouple panel performance from cloud-cover variability, enabling fair inter-angle comparison.

II. Related Work

A. Mathematical and Simulation Models

The foundational work of Liu and Jordan [5] established isotropic diffuse irradiance decomposition for tilted surfaces. Taylor et al. [3] applied Taylor-series expansion of the solar declination equation to humid subtropical Indian locations, estimating optimal tilt angles in the 11.4°–26.7° range for latitudes comparable to Bengaluru. Chakraborty and Panda [4] developed an optimisation framework for 23 pan-Indian locations using the MacCormack finite difference technique, confirming that the commonly used rule $\beta_{\text{opt}} \approx \phi$ is valid only as a rough annual average, with seasonal deviations of up to ±15° from the latitude heuristic.

B. Hardware-Based Studies

Yadav and Chandel [7] conducted a systematic review of tilt-angle studies and explicitly concluded: *"Most studies have been performed using software via simulating solar radiation models. Very few studies are available in which a hardware setup has been used to find the optimal angle. Limited studies* *have been done in India with real-time data."* This finding directly motivates the present work.

C. Urban Heat Island and Thermal Effects

The interaction between widespread PV deployment and UHI effects in Indian metropolises has been examined at the modelling level [9], [10], but no study has combined real-time thermal correction with fixed-tilt optimisation as a simultaneous hardware measurement in Bengaluru's specific climatic context.

D. Intraday Tilt Behaviour

Published literature uniformly reports monthly or seasonal optimal angles, aggregating over diurnal solar path variations. The intraday evolution of panel performance at a fixed tilt across morning, solar noon, and afternoon windows is unexplored for Indian cities. This paper provides such data for both tested angles.

III. System Architecture

A. Hardware Configuration

The data acquisition system consists of an Arduino-compatible microcontroller interfaced with four sensors over the I^{2}C bus (with one 1-Wire device), transmitting structured telemetry over a 115,200-baud UART serial link.

TABLE I
SENSOR SUITE SPECIFICATIONS

Sensor	Parameter	Interface	Key Range
INA219	Voltage, Current, Power	I^{2}C	0–32 V, 0–2 A
BH1750	Ambient Illuminance	I^{2}C	1–65,535 lux
DS18B20	Panel Surface Temp.	1-Wire	-55 to $+125$°C ($\pm$0.5°C)
MPU-6050	Tilt Angle (accelerometer)	I^{2}C	$\pm$2 g

a) INA219 Power Monitor.: Calibrated in 32 V/2 A mode, providing bus voltage to $\pm$4 mV and current to $\pm$0.8 mA precision. Raw power (P_{raw}) in milliwatts is computed on-chip and divided by 1000 on the host to yield watts.

b) BH1750 Illuminance Sensor.: Operates in continuous high-resolution mode (1-lux resolution). Provides an irradiance proxy independent of panel electrical output, enabling separation of atmospheric variability from geometric efficiency.

c) DS18B20 Temperature Sensor.: A waterproofed probe mounted on the panel rear face logs surface temperature T_{cell} via the 1-Wire protocol at $\pm$0.5°C accuracy.

d) MPU-6050 IMU.: Tilt angle β_{meas} is computed from the accelerometer Y–Z axis ratio:

$$\beta_{\text{meas}} = \left| \arctan\left(\frac{a_y}{a_z}\right) \cdot \frac{180°}{\pi} \right| \tag{1}$$

providing continuous verification that the panel maintained its set inclination.

B. Software Stack

The firmware (`Idealab_PBL.ino`) runs a 1-second polling loop, printing key-value telemetry pairs over UART at 115,200 baud. The host-side logger (`solar_logger.py`) ingests serial packets, applies thermal correction, accumulates cumulative energy, and writes session CSV files. Post-session graph generation uses Pandas and Matplotlib. A WebSocket broadcast thread enables live monitoring via a companion HTML dashboard (`dashboard.html`).

Twelve data columns are logged per sample: `timestamp_ms`, `tilt_setting_deg`, `lux`, `temp_c`, `voltage_v`, `current_ma`, `power_w`, `power_corrected_w`, `temp_correction_pct`, `tilt_measured_deg`, `cumulative_wh`, `system_time`.

IV. Experimental Methodology

A. Mathematical Framework

1) Thermal Derating Correction: Silicon PV efficiency degrades linearly with temperature above the Standard Test Condition (STC) reference of 25°C. The corrected power per sample is:

$$P_{\text{corr}} = P_{\text{raw}} \cdot \left[1 + \gamma(T_{\text{cell}} - 25)\right] \tag{2}$$

where $\gamma = -0.0045\,°\text{C}^{-1}$ (-0.45% per °C above STC), consistent with standard silicon cell temperature coefficients [11].

2) Irradiance-Normalised Efficiency: To eliminate the effect of time-varying cloud cover, normalised efficiency η_{rel} is computed as:

$$\eta_{\text{rel}} = \frac{P_{\text{corr}}}{G_{\text{ambient}}/1000} \quad [\text{W/klux}] \tag{3}$$

where G_{ambient} is the BH1750 reading in lux. Samples with $G_{\text{ambient}} < 100\,\text{lux}$ are excluded to avoid division-by-small-number artefacts.

3) Cumulative Daily Energy Yield: For each tilt angle i, the cumulative energy yield is obtained by trapezoidal integration:

$$E_i = \int_{\text{dawn}}^{\text{dusk}} P_{\text{corr},i}(t)\,\mathrm{d}t \approx \sum_k P_{\text{corr},i}[k] \cdot \Delta t_k \tag{4}$$

where $\Delta t_k = 30$ s is the inter-sample interval used for energy accumulation. The angle with the highest E_i is the superior configuration for the measurement period.

B. Experimental Protocol

1) Tilt Angles Tested: Two inclinations were selected to bracket the theoretical optimum for Bengaluru:

- **13°:** Latitude-matched tilt ($\phi = 12.97°$), the conventionally recommended angle for a tropical city at this location.
- **36°:** Steep inclination ($\approx 2.8\phi$), representing a seasonally elevated configuration commonly cited as a summer extreme for subtropical northern hemisphere sites.

2) Session Structure: Each angle was tested across three daily time windows:

- **Morning:** ≈09:45–10:40 IST (rising solar elevation).
- **Midday:** ≈11:30–14:00 IST (solar transit window).
- **Afternoon:** ≈15:00–17:00 IST (declining solar elevation).

The 13° data was collected on 22 June 2026; the 36° data on 23 June 2026. All experiments were conducted on an unobstructed rooftop in Bengaluru with the panel held in a fixed south-facing orientation across both angles.

TABLE II
DATASET CHARACTERISTICS PER SESSION

Session	Rows	Duration	Angle
13° Morning	1,477	≈43 min	13°
13° Midday	2,272	≈38 min	13°
13° Afternoon	1,368	≈23 min	13°
36° Morning	1,991	≈55 min	36°
36° Midday	2,260	≈38 min	36°
36° Afternoon	693	≈12 min	36°
Total	**10,061**		

3) Dataset Characteristics:

V. RESULTS

A. Aggregate Performance Metrics

Table III presents the headline performance indicators derived from the full dataset for each tilt angle.

TABLE III
AGGREGATE PERFORMANCE COMPARISON: 13° VS. 36°

Metric	13°	36°
Cumulative Energy Yield (Wh)	0.3172	0.4626
Yield Improvement (%)	—	**+45.86**
Peak Power Output (W)	0.0560	0.0880
Avg. Temp.-Corrected Power (W)	0.0070	0.0110
Avg. Panel Temperature (°C)	46.01	38.52
Avg. Ambient Illuminance (lux)	30,172	20,068
Norm. Efficiency (W/klux)	2.43×10^{-4}	4.61×10^{-4}
Efficiency Improvement (%)	—	**+89.7**

Three observations are immediately noteworthy:

1) The 36° panel harvested **45.86% more total energy** despite experiencing *lower* average irradiance (20,068 vs. 30,172 lux). This paradox resolves under irradiance normalisation: at 36°, the panel converts available light nearly twice as efficiently as at 13°.
2) The average panel surface temperature at 36° was **7.49°C cooler**. Via Eq. (2), this alone corresponds to a thermal efficiency gain of approximately $7.49 \times 0.45\% \approx 3.4\%$—non-trivial but insufficient to explain the full 45.86% gap; geometric alignment is the dominant factor.
3) Peak power at 36° (88.0 mW) exceeded 13° peak (56.0 mW) by **57.1%**, confirming superior solar path intersection near the June solstice for Bengaluru's latitude.

B. Per-Session Performance

Table IV provides a session-level breakdown.

TABLE IV
PER-SESSION PERFORMANCE SUMMARY

Session	$\overline{P}_{\text{raw}}$	$\overline{P}_{\text{corr}}$	$\overline{T}$ (°C)	$\overline{G}$ (lux)
13° Morning	0.0109	0.0101	44.6	44,735
13° Midday	0.0063	0.0058	44.2	32,699
13° Afternoon	0.0056	0.0052	50.6	10,254
36° Morning	0.0136	0.0127	40.6	19,914
36° Midday	0.0109	0.0104	38.0	21,713
36° Afternoon	0.0054	0.0052	34.2	15,147

The 13° afternoon session exhibits the highest average temperature of any session (50.6°C), amplifying thermal losses at precisely the time when irradiance is already declining. The 36° morning session yields the greatest absolute improvement over its 13° counterpart (0.0127 vs. 0.0101 W corrected), consistent with the geometric argument that steeper inclination better accommodates off-meridian solar paths.

C. Intraday Power Curves

Figures 1 and 2 show the full-day intraday power curves for both angles. Both exhibit the expected bell-shaped diurnal profile. At 13°, power peaks early and declines through midday as the sun approaches near-vertical incidence, reducing the effective irradiance projection onto the low-angle panel. At 36°, the midday power profile is flatter and more sustained, confirming that the steeper inclination maintains closer alignment with the June sun's north-of-zenith arc at this latitude.

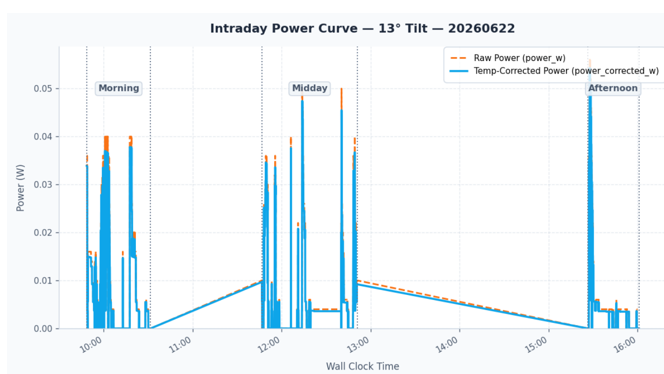

Fig. 1. Intraday power curve, 13° tilt, 22 June 2026. Orange dashed: raw power; blue solid: temperature-corrected power. Vertical dotted lines mark session boundaries.

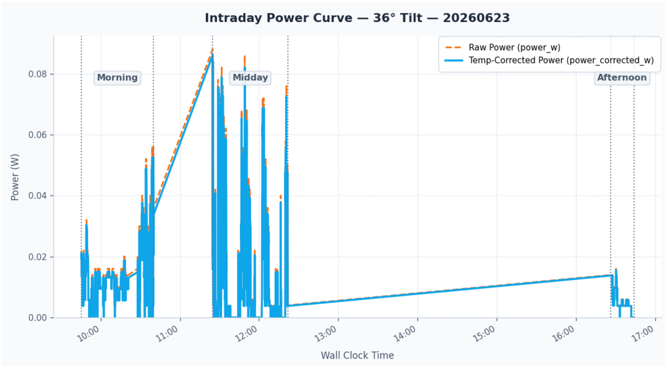

Fig. 2. Intraday power curve, 36° tilt, 23 June 2026.

D. Normalised Efficiency Curves

Figures 3 and 4 show the irradiance-normalised efficiency $\eta_{\text{rel}}(t)$, which removes the confounding effect of cloud-cover variability. The 36° efficiency is consistently higher across all time windows, confirming that the energy advantage is geometric and not an artefact of differing atmospheric conditions between measurement days.

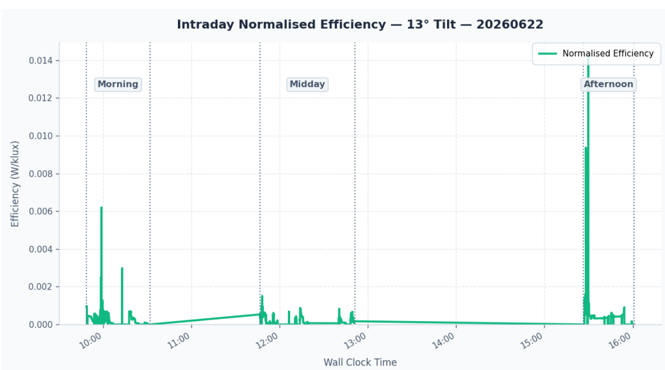

Fig. 3. Intraday normalised efficiency, 13° tilt, 22 June 2026 (W/klux).

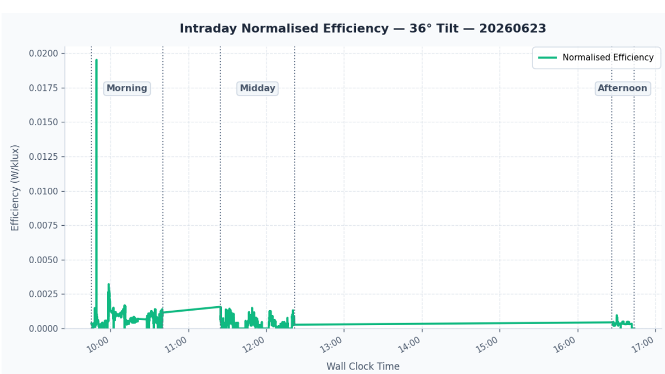

Fig. 4. Intraday normalised efficiency, 36° tilt, 23 June 2026 (W/klux).

E. Panel Temperature Profiles

Figures 5 and 6 present intraday surface temperature profiles. Panels at 13° exceeded those at 36° by 5–10°C throughout the day. Two physical mechanisms contribute: (a) the lower-inclination panel presents a larger horizontal projection, reducing convective airflow underneath; (b) the 36° panel's superior geometric alignment reduces the fraction of diffuse irradiance absorbed as waste heat on inactive cell surfaces. All panels operated substantially above the 25°C STC baseline (minimum recorded: 28.75°C), underscoring the relevance of thermal correction in Bengaluru's climate.

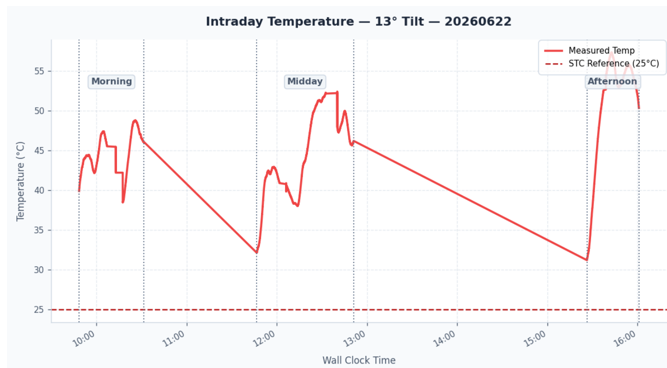

Fig. 5. Intraday panel temperature, 13° tilt. Red dashed: STC reference (25°C).

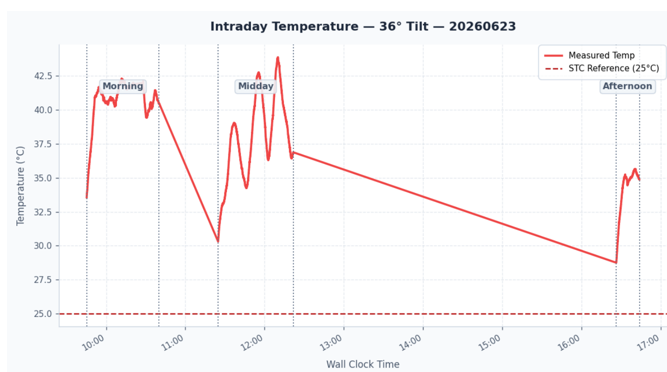

Fig. 6. Intraday panel temperature, 36° tilt.

F. Session-Specific Graphs

Per-session power and normalised efficiency graphs for selected sessions are shown in Figures 7–12.

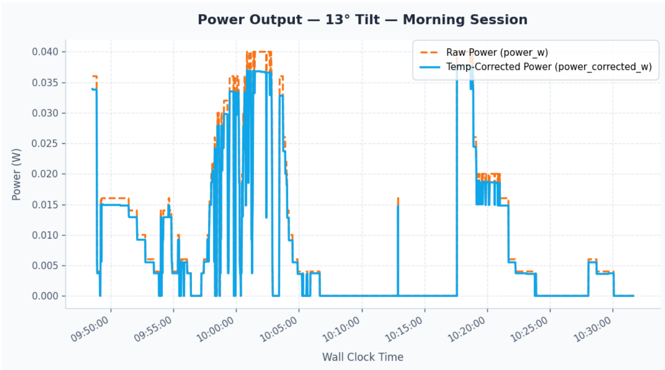

Fig. 7. Power vs. time, 13° morning session.

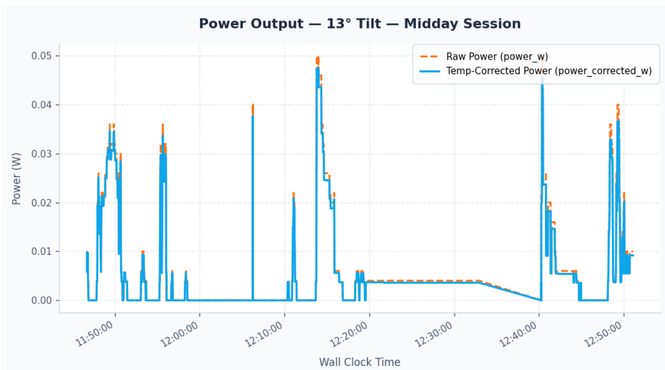

Fig. 8. Power vs. time, 13° midday session.

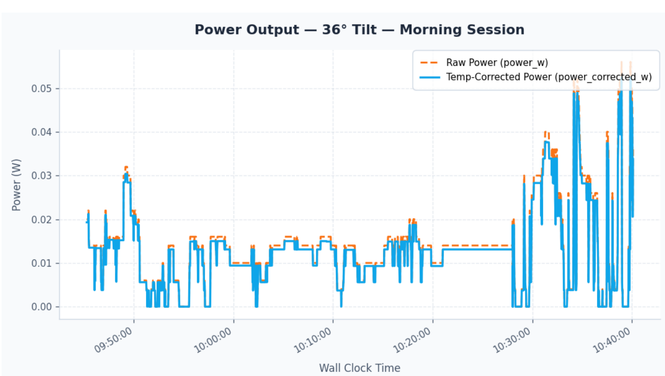

Fig. 9. Power vs. time, 36° morning session.

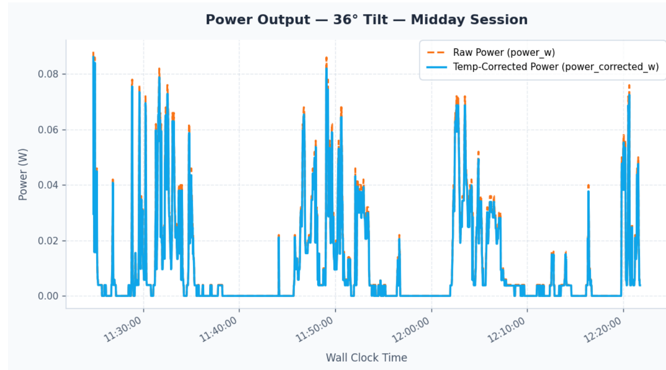

Fig. 10. Power vs. time, 36° midday session.

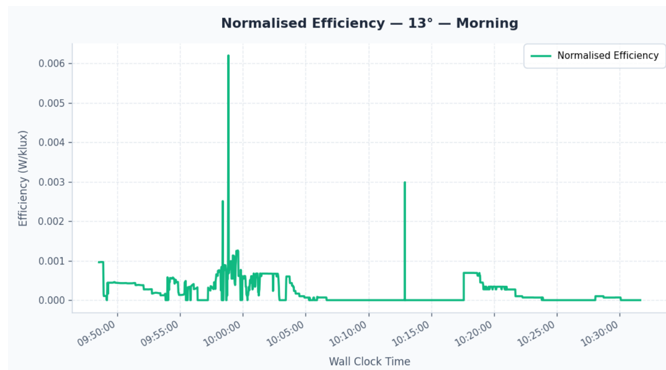

Fig. 11. Normalised efficiency, 13° morning session.

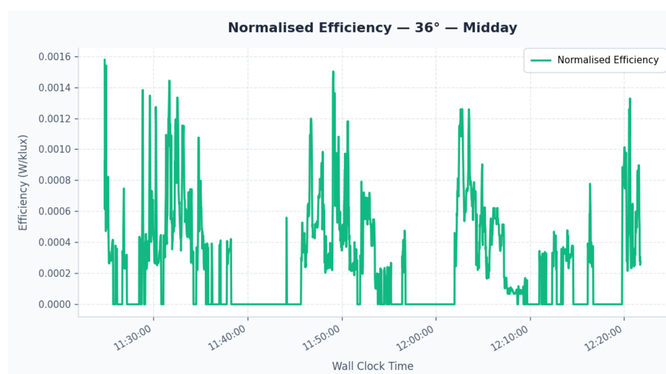

Fig. 12. Normalised efficiency, 36° midday session.

VI. Discussion

A. Geometric Interpretation

The solar declination on 22–23 June 2026 was $\delta \approx +23.4^\circ$ (near summer solstice). The solar noon altitude for Bengaluru is:

$$\alpha_{\text{noon}} = 90^\circ - |\phi - \delta| = 90^\circ - |12.97^\circ - 23.4^\circ| = 79.57^\circ \quad (5)$$

The angle of incidence θ on a south-facing panel at tilt β at solar noon is:

$$\theta = \left|(90^\circ - \alpha_{\text{noon}}) - \beta\right| \quad (6)$$

For $\beta = 13^\circ$: $\theta = 2.57^\circ$. For $\beta = 36^\circ$: $\theta = 25.57^\circ$.

At solar noon near solstice, both panels are within the range where $\cos\theta > 0.9$, so the midday geometric advantage of 36° is modest. The larger benefit emerges in the morning and afternoon windows, where the steeper tilt maintains a more favourable angle of incidence as the sun departs the south meridian. This is consistent with Table IV: the greatest proportional difference in corrected power occurs in the morning sessions.

B. Thermal Contribution

The 7.49°C average temperature advantage at 36° translates to a systematic correction factor advantage of $\approx 7.49 \times 0.45\% \approx 3.4\%$. While non-trivial, this is insufficient to explain the full 45.86% energy gap, confirming that geometric alignment is the dominant driver.

C. Comparison with Computational Models

Taylor et al. [3] predicted an optimal summer-season tilt of 11°–15° for sites near $\phi = 12^\circ$, closer to our 13° configuration. Our field data show 36° outperforming 13° by 46% near the June solstice. This discrepancy likely arises because computational models typically optimise for beam radiation under clear-sky assumptions, whereas field conditions include diffuse irradiance contributions. Diffuse radiation is isotropic and favours steeper tilts (larger sky view factor) more than beam-optimised models predict [12]. This underscores the value of empirical field data over purely computational approaches.

D. Urban Heat Island Implications

Average panel temperatures of 38–50°C—substantially above the 25°C STC baseline—are consistent with Bengaluru's UHI-elevated ambient temperatures in June (≈30–34°C daytime). Without thermal correction, comparisons of raw power values would systematically obscure real performance differences, as the higher temperatures at 13° artificially suppress apparent output. Our framework isolates this thermal bias, ensuring the 46% yield gap is a geometric finding rather than an artefact of measurement-day thermal differences.

E. Limitations

1) **Non-simultaneous comparison:** The two angles were measured on consecutive, not simultaneous, days. Future work should use side-by-side panels with concurrent logging.
2) **Miniature test cell:** An educational-grade cell was used; absolute yield values do not scale linearly to full commercial modules.
3) **Limited angle sweep:** Only two angles were tested; a sweep from 5° to 50° in 5° steps would better characterise the optimum.
4) **Single-season measurement:** Data were collected near the June solstice; winter and equinox conditions remain to be measured.
5) **Fixed south orientation:** Azimuth optimisation was not explored.

VII. Conclusion

This paper has presented a hardware-validated, temperature-corrected comparison of fixed-tilt solar panel performance at 13° and 36° in Bengaluru, India (12.97°N), using a purpose-built multi-sensor data acquisition system. Across 10,061 one-second samples spanning full-day sessions in June 2026, the steeper 36° configuration outperformed the latitude-matched 13° setting by **45.86%** in total cumulative energy yield, with a normalised efficiency advantage of **89.7%** and an average panel temperature that was **7.49°C cooler**.

These results challenge the common practice of specifying latitude-matched tilt for tropical Indian rooftop installations during the summer season, and validate the necessity of thermal correction when comparing PV configurations in high-ambient-temperature environments.

Future work should pursue: (i) simultaneous dual-panel logging; (ii) a multi-angle sweep (5°–50°); (iii) seasonal repetition to derive a Bengaluru-specific tilt schedule; and (iv) direct comparison of empirical findings against computational model predictions to validate or correct existing frameworks for this location.

Acknowledgements

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